MAIS PAC

IPhO 02.15: Thermodynamics and the Ideal Gas · Lecture + Problems

Thermodynamics and the Ideal Gas

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

5 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. What thermodynamics is about

Mechanics follows one particle; thermodynamics follows 10^(23) of them, and refuses to track them individually. Instead it uses a handful of **state variables** — pressure p, volume V, temperature T, internal energy U, entropy S — that depend only on the present state of the gas, not on how it got there.

The equation of state of an ideal gas ties three of them together:

$$pV = nRT = Nk_BT,\ R = 8.31\ \{J,mol^{(-1)},K^{(-1)}\},\ k_B = 1.38\times10^{(-23)}\ \{J,K^{(-1)}\}. \tag{1}$$

An ideal gas is one whose molecules have negligible volume and no forces between them except during collisions. Real gases approximate it well at low pressure and high temperature, and depart badly near condensation.

Temperature must be on the absolute (kelvin) scale in every gas equation. Absolute zero is where the pressure of a fixed amount of gas would extrapolate to zero: -273.15°{C}.

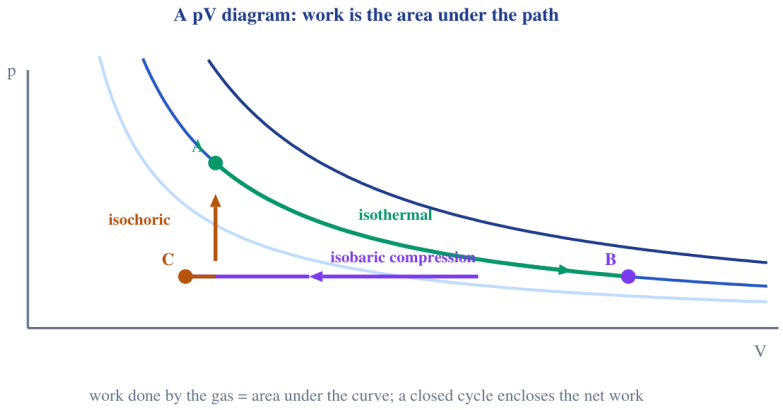


Figure 1. The area under a path is the work done in that process, and the area enclosed by a cycle is the net work. The direction of travel matters: clockwise means the engine does work on the outside world.

2. Kinetic theory: pressure from molecular collisions

Each collision with a wall transfers momentum, and a vast number of collisions per second constitutes a steady pressure. The result of the calculation is

$$p = (1)/(3)(Nm)/(V)\{v^2\} = (1)/(3)\rho,\{v^2\}, \tag{2}$$

and comparison with pV = Nk_BT identifies

$$(1)/(2)m\{v^2\} = (3)/(2)k_BT. \tag{3}$$

That is a profound statement: **temperature is the average translational kinetic energy per molecule**, and it is independent of the gas species. The three characteristic speeds differ by small numerical factors:

$$v_{\{rms\}} = \sqrt{((3k_BT)/(m))} = \sqrt{((3RT)/(M))},\ v_{\{mp\}} = \sqrt{((2k_BT)/(m))},\ v = \sqrt{((8k_BT)/(\pi\, m))}. \tag{4}$$

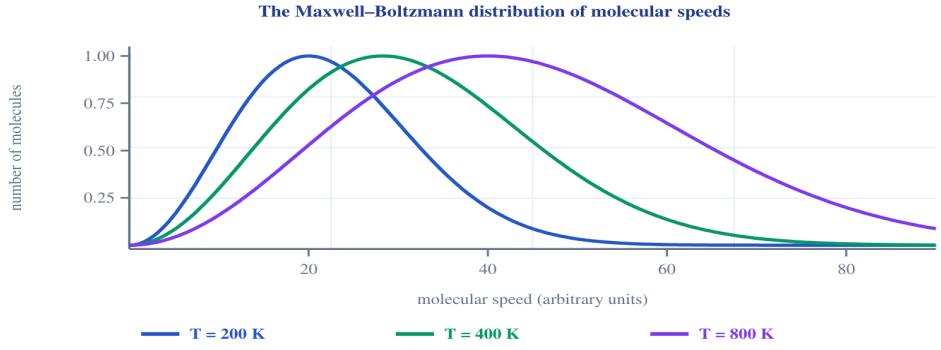


Figure. Raising the temperature shifts the peak to higher speed; each curve is scaled to its own peak to compare shapes.

Interactive experiment: The ideal gas law — open the online handout to use it.

The distribution of speeds is the Maxwell–Boltzmann distribution, with a long tail at high speeds. That tail is why evaporation happens below the boiling point, why chemical reactions have a temperature-dependent rate (the Arrhenius law), and why a gas at 300 {K} still contains a few molecules fast enough to escape the Earth's gravity — over geological time, that is how the atmosphere lost its hydrogen and helium.

3. The first law: energy bookkeeping

$$\Delta U = Q - W,\ W = \int p,dV. \tag{5}$$

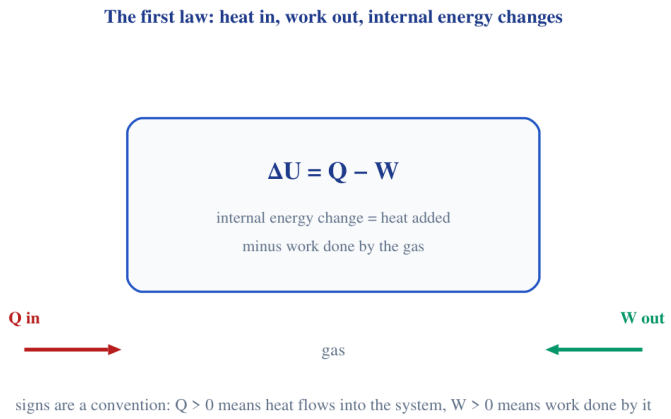


Figure 2. The first law is just energy conservation with a sign convention. The second law, which is not about energy at all, says that not all of Q can be turned into W.

The internal energy of an ideal gas depends only on T: U = (f)/(2)nRT with f the number of degrees of freedom (3 for monatomic, 5 for diatomic at ordinary temperatures). Then the four standard processes:

process	condition	work	heat	result
isothermal	T const	nRTln (V_2/V_1)	Q = W	Δ U = 0
isobaric	p const	p,Δ V = nRΔ T	C_p nΔ T	W = nRΔ T
isochoric	V const	0	C_V nΔ T	Δ U = Q
adiabatic	Q = 0	-Δ U	0	pV^γ = const

The ratio γ = C_p/C_V = 1 + 2/f is 5/3 for a monatomic gas and 7/5 for a diatomic one — measurable in the speed of sound, and the reason a bicycle pump gets hot.

Two useful consequences of the adiabatic relation: a gas expanding adiabatically cools (the basis of refrigeration and of the condensation trail behind a wing), and a sound wave is an adiabatic compression, so v_({sound}) = √(γ p/ρ).

4. The second law and engines

The first law forbids perpetual motion machines of the first kind (creating energy). The second forbids the second kind: you cannot convert heat completely into work in a cyclic process.

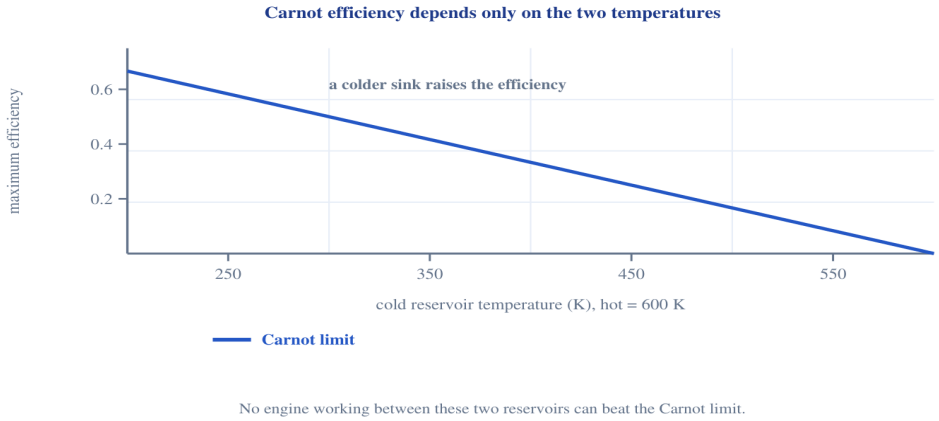


Figure. No engine working between these two reservoirs can beat the Carnot limit.

The **Carnot efficiency** is the absolute ceiling for any engine working between a hot reservoir at T_H and a cold sink at T_C:

$$_({Carnot}) = 1 - (T_C)/(T_H), = (W)/(Q_H). \tag{6}$$

Real engines (steam turbines, car engines) reach perhaps 40% of what the Carnot limit allows at their temperatures; the limit is set by the temperatures, not by engineering cleverness.

Entropy is the quantitative version of the second law:

$$dS = (dQ_({rev}))/(T), \Delta S_({universe}) 0. \tag{7}$$

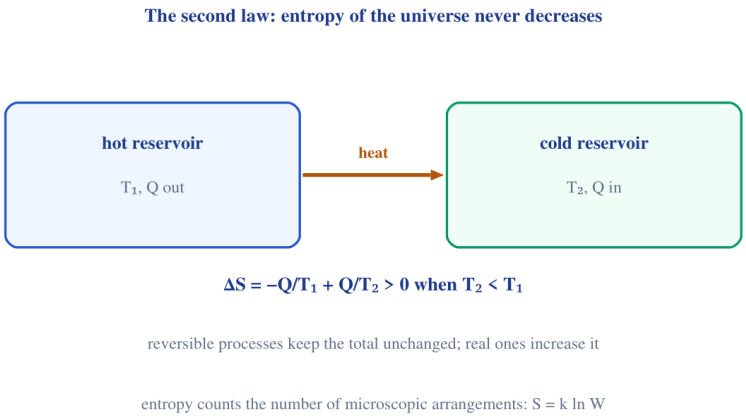
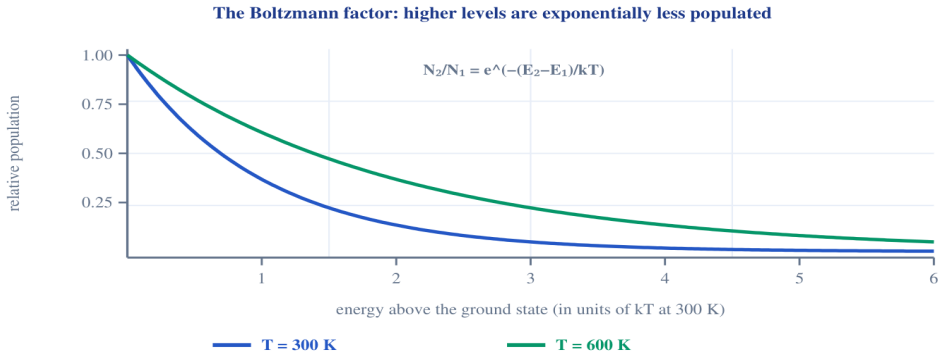


Figure 3. Heat flows spontaneously from hot to cold, and the entropy increase is the quantitative statement of that arrow of time. An engine between the reservoirs can extract work, but the total entropy still rises.

For an ideal gas, $\Delta S = nC_V \ln(T_2/T_1) + nR \ln(V_2/V_1)$, which shows that free expansion into a vacuum (where T is unchanged but V grows) increases entropy — a process that is irreversible though not energy-changing. Boltzmann's formula

$$S = k_B \ln W \tag{8}$$

identifies entropy with the number of microscopic arrangements consistent with the macroscopic state, which is why the second law is statistical: it is not that a shattered glass **cannot** reassemble, only that the number of ways for it to stay shattered is astronomically larger.



At low temperature almost every particle sits in the ground state; at high temperature the levels fill up.

Figure. At low temperature almost every particle sits in the ground state; at high temperature the levels fill up.

The thermodynamics checklist. Convert to kelvin at once → decide which quantity is held constant, then use the right process (isothermal, isobaric, isochoric, adiabatic) → for anything with work, sketch the pV diagram and shade the area → for kinetic theory, (1)/(2)m{v²} = (3)/(2)k_BT links the microscopic picture to p and T → for engines, compute = W/Q_H and compare with 1 - T_C/T_H → for entropy, count states, not energies.

5. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — The ideal gas law. A cylinder contains 0.25 {mol} of ideal gas at 27°{C} and a pressure of 1.5×10^5 {Pa}. (a) Find the volume. (b) Find the number of molecules. (c) Find the new pressure if the gas is compressed to half the volume at constant temperature.

Worked solution →

[1] Problem 2 — Molecular speeds. Find the rms speed of (a) helium atoms and (b) nitrogen molecules at 300 {K}, taking M_({He}) = 4.0 {g,mol^(-1)} and M_({N_2}) = 28 {g,mol^(-1)}. (c) Find the temperature at which the rms speed of nitrogen equals that of helium at 300 {K}.

Worked solution →

[2] Problem 3 — Work in three processes. One mole of monatomic ideal gas expands from V_1 = 10 {L} to V_2 = 20 {L} at T = 300 {K}. (a) Find the work done if the expansion is isothermal. (b) Find the work if it is isobaric starting from the same initial state, and the final temperature. (c) Explain why the answers differ and which process needs more heat.

Worked solution →

[2] Problem 4 — An adiabatic compression. A monatomic ideal gas at 300 {K} and 1.0×10^5 {Pa} is compressed adiabatically to one eighth of its volume. (a) Find the final pressure. (b) Find the final temperature. (c) Find the work done on the gas if there are 2.0 {mol}.

Worked solution →

[2] Problem 5 — A Carnot engine. A heat engine operates between 500 {K} and 300 {K}, absorbing 2.0 {kJ} per cycle from the hot reservoir. (a) Find the Carnot efficiency and the maximum work per cycle. (b) Find the heat rejected per cycle. (c) Find the entropy change of the universe per cycle for a real engine with an efficiency of 30%.

Worked solution →

[2] Problem 6 — Kinetic theory estimates. Air at 300 {K} and 1.0×10^5 {Pa} has molecules of mass 4.8×10^(-26) {kg} and diameter about 3.0×10^(-10) {m}. (a) Find the number density. (b) Estimate the mean free path. (c) Estimate the collision frequency and compare it with the rms speed.

Worked solution →

[2] Problem 7 — Mixing and entropy. Two identical insulated containers hold one mole each of the same ideal gas at the same temperature, one at volume V and one at volume 2V. The partition between them is removed. (a) Find the final temperature and pressure. (b) Find the entropy change of the gas. (c) Explain why the process is irreversible even though no energy is exchanged.

Worked solution →

[3] Problem 8 — A cyclic process. One mole of monatomic gas goes through a cycle: isothermal expansion at 400 {K} from 10 {L} to 20 {L}, isochoric cooling to 300 {K}, then isothermal compression back to 10 {L}, then isochoric heating back to 400 {K}. (a) Find the net work done per cycle. (b) Find the heat exchanged in each step. (c) Find the efficiency and compare with the Carnot efficiency between the same two temperatures.

Worked solution →

[3] Problem 9 — The speed of sound.

(a) Show that for an ideal gas the speed of sound is $v = \sqrt{\gamma RT/M}$. (b) Evaluate it for air at 20°{C}, with $\gamma = 1.40$ and $M = 0.029$ {kg,mol^(-1)}. (c) Explain why the speed of sound does not depend on pressure at fixed temperature, and why it *does* depend on humidity.

Worked solution →

[3] Problem 10 — A real engine. A car engine takes in air at 300 {K} and burns fuel at 2200 {K}, and rejects heat to the surroundings at 400 {K}. (a) Find the Carnot efficiency for these temperatures. (b) Find the actual efficiency if the engine produces 60 {kW} while burning fuel at a rate of 200 {kW}. (c) Explain the main reasons the real efficiency is far below the Carnot limit.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

MIT OpenCourseWare 8.01/8.02 thermodynamics notes

Kinetic theory, the first and second laws, and heat transfer worked examples.